Clinical Study Data Reviewer's Guide (cSDRG)

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Field	Value
Document	Clinical Study Data Reviewer's Guide (cSDRG)
Study	TROPIC / EFC6193 / NCT00417079
Compound	Cabazitaxel (CbzP) vs Mitoxantrone (MP)
Standard (package layer)	CDISC SDTMIG v3.4 + CDISC/NCI CT 2026-03-27 (uplifted; Section 5)
Standard (pristine source)	CDISC SDTMIG v3.1.1 (PDS 2013; local 01_source_data/real_sdtm/)
Document version	2.0
Effective	2026-08-12
Supersedes	SDRG v1.3 from the v0.2.2-portfolio release
Product claim	Controlled clinical-submission simulation; not a regulatory submission

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0. What this package is / is not (read first)

This package is	This package is not
Study-data explanation for a submission-style Module 5 tree	An FDA filing / real application sequence
Real MP-only de-identified SDTM (N=371) as source for dual-language ADaM	Two-arm real patient-level IPD in git
Uplifted SDTMIG 3.4 package layer + define co-located under m5/.../tabulations/sdtm/	Proof that commercial P21 cleared every residual
CORE SDTM 3.4 run with classified residuals (F-015)	"Full CORE clean" / zero-finding claim
Week-offset / partial-date source honesty (F-017)	Day-level AE timing precision that the source never had
Controlled simulation with patient XPT not redistributed	Part 11 validated system

Binding claim: docs/PRODUCT_CLAIM.md

Source intake pack (WS-1): docs/workstreams/WS1_SOURCE_INTAKE_PACK.md

Residual risks: docs/workstreams/WS5_KNOWN_DIFFERENCES_MEMO.md

External validation slots: docs/workstreams/WS3_EXTERNAL_VALIDATION_EVIDENCE_INDEX.md

Current controlled release: tag v0.3.0-clinical-simulation docs/RELEASE_NOTE_v0.3.0-clinical-simulation.md python3 scripts/verify_release.py

Review package face: 08_submission_package/m5/datasets/tropic/tabulations/sdtm/

Analysis narrative: ADRG.md

Findings disposition: 06_qc_evidence/audit/FINDINGS_DISPOSITION_BOARD.md

Redistribution: Real .sas7bdat and package .xpt are not in Git (data rights and repository surface policy). Structure, Define-XML, cSDRG/ADRG, and programs are retained.

0A. CRF grounding (principal source-fidelity rule)

CRF PDF (source): 01_source_data/Sanofi CRF Tropic.pdf (EFC6193, FINAL 21-Nov-2006)

In package: blankcrf.pdf under tabulations/sdtm (source CRF copy -- not claimed as a complete annotated aCRF with page-level define origins)

Decision record: docs/workstreams/reviews/WS1_CRF_GROUNDING_D012_2026-07-09.md

Rule

Do	Do not
State what the form collected	Imply "trial never collected X" when the CRF has a field for X
State what PDS retains (and at what precision)	Blame study design for public-extract reductions
Split gaps: CRF non-collection vs extract reduction vs programming scope	Invent values for null/stripped fields

Domain grounding summary (audit D-012)

Domain	CRF form (examples)	What sponsor collected	What PDS extract retains	Limitation class
AE	Adverse Event Form O.1_AE_1	Diagnosis; status; grade; relationship to study treatment; action taken; corrective therapy; outcome (incl. fatal); seriousness Y/N + seriousness criteria	Terms/MedDRA; AETOXGR (incl. grade 5 fatal); AEREL/AEACN/AECONTRT/AEOUT largely populated; AESER + criteria flags largely present	Class A (core safety retained). Class B for date precision (CRF day/month/year -> PDS week offsets AESTWK/AEENWK). Residual blank AESER on a minority of rows = null/incomplete, not "column never collected"
LB	Hematology LABH_1, Biochemistry LABB_1, PSA	CBC panel; electrolytes (Na/K/Cl/bicarb); LFTs; creat/BUN; glucose; testosterone; PSA	Matching LBTESTCD set including SODIUM,K,CL,BICARB, ALP,CREAT,PSA, etc.	Class A -- electrolytes are on CRF and in extract (not a TROPIC "electrolyte never collected" story)
ECOG	VS panel ECOG 0-4	Performance status 0-4	VSTESTCD=ECOG with observed 0-4	Class A

Domain	CRF form (examples)	What sponsor collected	What PDS extract retains	Limitation class
DS	End of Treatment O.ENDTT_2; End of Study O.ENDST_1	Main reason for stopping treatment/study (progression, AE, completed, LTFU, subject request, death, other, ...)	DSCAT/DSSCAT/DSDECOD/DSTERM carry those concepts	Class A for reason concepts present in extract

Explicit non-claims: full aCRF annotation package; day-true AE onset dates in public extract; commercial filing CRF provenance.

Package domain scope (analysis-scoped SDTM)

Layer	Content
Pristine PDS (local)	34 .sas7bdat domains (incl. EG, MH, PE, SV, IE, CD, CX, trial design fragments, etc.)
Package / define_sdtm.xml	18 datasets: DM, EX, AE, LB, CM, DS, VS, LS, PN + SUPPAE/CM/DM/DS/EX/LB/LS + TA, TS

Domains present in PDS but not packaged (by design for this controlled analysis scope):

CD, CX, EG, IE, MH, PE, PR, SC, SV, TE, TI, TV (+ related SUPP).

Do not claim the Module 5 SDTM folder is a full copy of every PDS domain.

Full E2E audit: docs/workstreams/reviews/WS1_SDTM_E2E_AUDIT_2026-07-09.md

Material extract residuals (SDTM E2E)

ID	Finding	Honesty
F-028	One subject has EXTRT=XRP6258 (10 rows) while all DM rows are ARM=MITOXANTRONE/PREDNISONE	Arm authority = DM/ADSL, not EXTRT alone; do not silently re-code EX
AE coverage	AE subjects 357/371 (14 subjects with no AE rows)	Any-AE rates must use ADSL denominator
F-026	~1134 BASELINE AE skeleton rows (blank AESER)	TEAE analyses use TRTEMFL

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1. Source data normalization & integrity controls

Source data are the official Sanofi de-identified SDTM for NCT00417079 (2013), accessed via Project Data Sphere (PDS) as SAS .sas7bdat. Provenance: root README 00_governance/REPRODUCIBILITY.md.

Staging: L_staging_ingest.sas (and R twin v_staging_ingest.R) ingests realsdtm.<domain>, transpose-merges SUPP--, and coerces character continuous indicators to numeric where required.

Critical single-arm source limitation: PDS public data here is MP only (N=371). Cabazitaxel is not present as source SDTM. Dual-language ADaM reconciliation is MP-only. Synthetic/reconstructed CbzP is merged only at TFL reporting (tfl_generation.R) -- non-confirmatory (F-003). Do not describe this as a two-arm double-programming of trial IPD.

Database write-protection architecture

- realsdtm libref -> 01_source_data/real_sdtm/ with access=readonly in 00_config.sas
- staging libref -> 04_analysis_datasets/staging/, writable and fully regenerable for SUPP-merged intermediates during SAS/ODA runs; the R twin writes domain .rds files to the same governed staging zone
- Final ADaM products write under 04_analysis_datasets/adam/; no program writes derived intermediates into the immutable source directory

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2. SDTM Domain Mapping Summary

Standard SDTM mapping structures were built in S_sdtm_mapping.sas from trial-era SDTM-IG 3.1.1 source data; SAP v4.0 treats this as source provenance and requires the release package to use the governed SDTMIG 3.4 uplift layer:

- DM (Demographics): Unique subject identifier USUBJID constructed via STUDYID || '-' || SITEID || '-' || SUBJID. Randomization date RANDDT and treatment start date TRTSDT mapped to standard ISO 8601 date fields.
- EX (Exposure): Normalised cycle-level actual administered doses (EXDOSE in mg).
- AE (Adverse Events): Coded utilizing MedDRA dictionaries into AEDECOD, AEBODSYS, and CTCAE-style grades (AETOXGR). CRF grounding (D-012): the Adverse Event form collected seriousness, relationship to study treatment, action taken, outcome (incl. fatal), and seriousness criteria -- and those concepts are largely present in the PDS extract (AESER, AEREL, AEACN, AEOUT, AESxxx flags). Do not narrate "trial never collected seriousness." Date Precision Note (Class B reduction): CRF collected calendar day/month/year; the public extract commonly carries week-offset integers (AESTWK, AEENWK). When a complete date is unavailable, analysis dates are reconstructed as RFSTDTC + (week - 1) 7 and retain the corresponding week-precision limitation. ADSL death dating preferentially uses a complete source AEDTHDTC when available and uses the DS week reconstruction only as fallback. This is an extract design / de-identification property, not a programming artefact.
- LB (Laboratory): Mapped continuous Absolute Neutrophil Count (ANC) and Prostate Specific Antigen (PSA) measurements.
- DS (Disposition): Captured study completion reasons, trial exits, and survival follow-up records.
- Disposition-derived clinical signal: DS progression records may be retained as a separately typed CLINPROG analysis signal, and DS death records support survival follow-up. CLINPROG is not a RECIST assessment and does not feed BOR, ORR, TTUMOR, or PFS.

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3. Reference Ranges & Baseline Criteria

- ADSL and ADLB laboratory baselines use deterministic source records flagged LBBLFL='Y'; they are not replaced by an arbitrary missing-day or population-constant baseline.
- Normal reference ranges (LBNRLO and LBNRHI) were preserved from raw PDS metadata. Lab values outside these ranges are flagged accordingly in LBNRIND.

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4. Known Data Limitations & Derivation Decisions

4.1 Baseline Laboratory Missingness

ADSL carries observed PSABL, ALPBL, and HGBBL values from source records flagged LBBLFL='Y'; a missing source baseline remains missing. No population-median proxy is inserted. ALBBL and LDHBL are unavailable in the public source release and remain missing with blank imputation flags. ECOGBLIF, PSABLIF, ALPBLIF, and HGBBLIF are 'N' because the real-data track performs no imputation.

These variables are not efficacy-model covariates. The primary and secondary Cox/log-rank analyses stratify only on pooled observed randomization factors (ECOGBL 0-1 vs 2 and MEASDISF; see 05_outputs/tfl/tfl_generation.R, compute_tte_stats() -> strata(ECOGBLGRP, MEASDISF)).

4.2 Supplemental Domain Ingestion

Domains LS (Lesion) and PN (Pain/Numeric) do not have supplemental (SUPPLS, SUPPPN) datasets in the PDS source data. The %transpose_supp() macro gracefully handles this via the supp_exists = 0 guard path, copying the primary domain directly without SUPP merge.

4.3 Country and Region Assignment

The DM domain in the source data does not contain country-of-study-site information. COUNTRY and REGION are not present in the de-identified release and are not derived (see the note in B_bimo_generation.sas); no placeholder geography is assigned. Geographic subgroup analyses are not part of SAP v4.0 reporting and are not reported.

4.4 Hardcoded Demographic Constant (SEX = 'M')

The demographics domain (DM) contains a hardcoded variable SEX = 'M' assigned to all subjects in A_adsl_generation.sas. This is a clinical decision consistent with the trial protocol for metastatic castration-resistant prostate cancer (mCRPC), which is an exclusively male patient population. To ensure metadata conformity, the Define-XML codelist references are maintained; however, no female subjects are present in the analysis dataset.

4.5 Partial/Imprecise Source Date Values (CM, LB, LS, PN)

The independent R SDTM validation (04_analysis_datasets/programs/r/v_sdtm_validation.log) raises four [WARNING]s flagging partial or imprecise ISO-8601 date values in source date fields: CMSTDTC (CM), LBDTC (LB), LSDTC (LS), and PNDTC (PN) -- e.g. ----07, --12-26, 2009---04. These are expected manifestations of the public-release date precision. Values are preserved rather than assigned fabricated day precision; derivations requiring a complete calendar date accept only parseable complete dates or use a separately governed fallback (for example the SV hierarchy in F-042). The warnings therefore remain visible and their downstream exclusions/fallbacks are reviewable.

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5. SDTMIG 3.4 Conformance Uplift (2026-06-20)

The pristine source SDTM (01_source_data/real_sdtm/, PDS 2013) was authored to SDTMIG 3.1.1, which is below the FDA Data Standards Catalog support floor. A derived, conformance-uplifted SDTM layer was produced to SDTMIG 3.4 + CDISC/NCI CT 2026-03-27 and is the version described by define_sdtm.xml and packaged in 08_submission_package/m5/.../tabulations/sdtm/. The raw source is never modified; the uplift is a deterministic derivation step (platform/uplift_sdtm_34.R for data, 03_metadata/define/uplift_define_34.py for the define).

Standard derivations applied (each preserves source data values):

- DM.AGE derived numeric from the de-identified AGEGRP (the PDS release masked exact age into AGEGRP; subjects coded >=85 are floored to AGE=85, with the cap flagged in SUPPDM QNAM=AGEGRP). The non-standard AGEGRP is removed from DM. ACTARM/ACTARMCD added (single completed arm, code A).
- AE.AESOC populated equal to AEBODSYS (MedDRA SOC already carried in AEBODSYS).
- EPOCH derived for AE/EX/VS (DS already carried it). Subject Elements (SE) are absent from the de-identified extract, so EPOCH is derived from the collected VISIT structure: SCREENING->SCREENING; BASELINE/CYCLE n/END OF TREATMENT->TREATMENT; FOLLOW-UP n->FOLLOW-UP; UNSCHEDULED->TREATMENT. All values are valid EPOCH CT.
- EX.EXENDY derived (study day of EXENDTC vs RFSTDTC).
- Week-offset timing (AESTWK/AEENWK/AESTWKF/AEENWKF, DSSTWK/DSSTWKF) -- non-standard in the parent domains -- relocated to SUPPAE/SUPPDS as supplemental qualifiers (linked by IDVAR/IDVARVAL), preserving the +/-3.5-day timing (see Section 2 Date Precision Note and the date-precision sensitivity analysis, 06_qc_evidence/audit/run_records/DATE_PRECISION_SENSITIVITY_2026-06-20.md).
- Redundant SUBJID removed from non-DM domains; non-standard ARM2/ARMA/ARMCD2 (define-only phantoms, never in data) dropped from IG.DM.
- TS enriched with public NCT00417079 parameters (NARMS=2, ACTSUB=371, SSTDTC=2007, AGEMIN=P18Y). TA (Trial Arms) built from the public two-arm design.
- Variable labels title-cased, leading/trailing whitespace stripped, variable order aligned to the CDISC SDTM library.

5.1 CORE SDTMIG 3.4 -- honesty (F-015)

Run record: platform/conformance/CORE_SDTM34_RUN_RECORD.md

Engine: cdisc-rules-engine (CORE) v0.16.0 standard SDTMIG 3.4 CT 2026-03-27

Headline: targeted structural uplift rules cleared; overall issue count is not zero and must not be marketed as "CORE clean."

Class	Examples	Disposition (controlled scope)
Structural targets fixed	AESOC, AGE, EPOCH, EXENDY, non-standard->SUPP, labels/order/type	Accept as uplift success
Inherent de-identification	SITEID / COUNTRY / MedDRA hierarchy codes / exact AE dates removed by PDS	Accept -- cannot invent PII or lost codes
Real source-data quality	AESER consistency (CORE-000266/022); VSSTRESC/N (CORE-000732)	Accept -- do not overwrite true safety source to greenwash

Class	Examples	Disposition (controlled scope)
Cross-domain N/A	RELREC/FAOBJ (CORE-000767) without FA in analysis-scoped package	Waive / N/A for this package scope
Engine-internal	CORE-000929 / CORE-001081 evaluation dataset failed to build	Accept with note -- tool noise, not silent data fix

Rule for reviewers:

- Do claim: "Uplifted package layer; CORE run recorded; structural targets cleared; residuals classified."
- Do not claim: "Full commercial conformance" or "zero CORE findings."
- Rule-level disposition matrix (WS-1): docs/workstreams/WS1_CORE_RESIDUAL_MATRIX.csv -- filed; still not a "CORE clean" claim.

Related residual: week-offset / partial ISO dates -- F-017 (Section 2 AE note, Section 4.5). Never invent day precision.

Pinnacle 21: Community 4.1.0 executed against the ADaM package using FDA engine 2508.1 and ADaMIG 1.3 (FDA). The run is informative only because Community is not Enterprise and the output reports Incompatible CLI used. Licensed final-package validation remains NOT_EXECUTED. See 06_qc_evidence/conformance/p21_adam_runrecord.md.